Ziyang Zhang

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Ph.D. student at Tsinghua EE. Research: time-series LLMs (understanding, forecasting, multimodal reasoning) and world models, speech generation, LLM alignment.

Education

Tsinghua University — Beijing, China
Ph.D. in Electronic Engineering (Advisor: Assoc. Prof. Chao Zhang; expected 2029) — *Aug. 2024 – Present*

Tsinghua University — Beijing, China
B.Eng. in Electronic Engineering; Outstanding Graduate (2024); Academic Excellence Scholarship ×3 — *Sept. 2020 – July 2024*

Research Internships

Shanghai Artificial Intelligence Laboratory — Nov. 2024 – Oct. 2026
Research Intern, Intern-S time-series team; all time-series and multimodal work below was done here

Intern-S2-Preview: A Scientific Agentic Foundation Model (397B) *Time-series modality lead* — Tech. report 2026

Motivation Handle longer, higher-frequency scientific signals and add forecasting: read the instruction in dialogue, decide the horizon and output future values.

Approach A long-sequence encoder (Q-Former compressive patching, then a channel-wise Transformer) raises the maximum input length from about 240k (previous Intern-S1-Pro, 1T parameters) to 300k time steps, with about 5–6× faster inference and about 20% of the GPU memory at maximum length. A new forecasting branch: a Q-Former gathers LLM semantics and encoder features to condition a causal Transformer forecaster via cross-attention, and a horizon module reads the instruction. On SciTS it beats Intern-S1-Pro on 7 of 9 shared understanding tasks (PHU01 F1 36.8 → 66.9); on all 7 forecasting tasks its MAPE is below our earlier TimeOmni, and on 6 below the best specialized time-series model; horizon accuracy is 99%.

My role Responsible for the time-series modality's overall architecture and training (merged into the main model); solely responsible, from scratch, for the architecture of the new forecasting capability.

Intern-S1: A Scientific Multimodal Foundation Model (241B MoE) *Time-series modality lead* — Tech. report 2025

Motivation Let a scientific foundation model read raw signals (seismic and gravitational waves, light curves, EEG; once a day to GHz rates) as a modality alongside images.

Approach Added a time-series encoder (adaptive downsampling + Transformer blocks) to an MoE model with 28B activated / 241B total parameters, continually pre-trained on 5T tokens, compressing raw signals into representations the LLM can reason over.

My role Responsible for the time-series modality's architecture and training: encoder design, cross-modal alignment with the LLM, and SFT; ran the merged training of the time-series modality into the 241B model on the lab's XTuner framework.

MMTA & TimeOmni-v: Multimodal Time-Series Analysis with Text and Vision *Co-first author (listed first)* — **NeurIPS 2026**

Motivation Most time-series benchmarks pair series with text or with plots of the same signal, so whether MLLMs can combine time series with real visual evidence was untested; e.g., judging whether a robot finished a task needs the instruction, egocentric video and joint trajectories together.

Approach Built MMTA, a tri-modal benchmark over 9 domains (robotics, driving, healthcare, etc.) in video+series and image+series layouts, with a 3,524-sample test split for classification and forecasting. Zero-shot tests of 8 MLLMs show a shared bottleneck: series written as text give very long token sequences and weak alignment with vision. Proposed TimeOmni-v, which adds a Chronos-2 time-series encoder and a forecasting head to Qwen2.5-Omni so it handles time series natively, with a time-series RoPE that puts series and video on one timeline and interleaves series tokens with video frames. Fine-tuned per dataset, it reaches an average classification swF1 of 0.601 (strongest zero-shot baseline, Gemini-3-Flash: 0.442); on the long-video robotics task AgiBot it uses 232.8× fewer time-series tokens than text input and runs 23× faster.

SciTS & TimeOmni: Scientific Time Series Understanding and Generation *Second author* — **ICLR 2026**

Motivation Scientific series reach ten million points and 10 MHz; as text they are too long, as images they lose precision. Goal: a benchmark for understanding and generation, and a model that treats time series as a modality of its own.

Approach SciTS covers 12 disciplines, 43 tasks and 54,023 samples; zero-shot tests of 17 LLMs, MLLMs and time-series models show that general LLMs generalize better than specialized time-series models. TimeOmni, a unified time-series LLM trained on SciTS, ranks first on average in both understanding and generation.

My role Solely responsible for TimeOmni's architecture and training: a router picks, by signal length, a patch-expert encoder (patch sizes 1–2048) so the encoded length stays at 100–200; encodings are reprogrammed into the LLM's space and a forecasting head outputs values; joint training of understanding and generation on Qwen3-8B.

TimeOmni-o1: Bidirectionally Coupling TSFMs and LLMs for Reasoning and Forecasting *First author* — Preprint 2026

Motivation Series often come with text such as event reports, and both understanding and forecasting need reasoning over the two. LLMs handle long series poorly; time-series foundation models (TSFMs) forecast well but cannot read or write text; time-series LLMs forecast on the LLM side and leave the TSFM's pretrained forecasting unused.

Approach Couples a pretrained TSFM (Chronos-2) and an LLM (Qwen3.5-9B) both ways: the TSFM passes a compressed encoding to the LLM, which reasons in text; the reasoning returns to the TSFM via gated cross-attention to condition quantile forecasts, end to end. Built MMTR (500k+ reasoning-annotated samples, 12 domains); SFT, then GRPO on understanding. On the MMTR test set, understanding averages 84.2 over 12 domains (best baseline 74.9); forecasting CRPS is 0.116, the lowest in all 5 domains and below its inner LLM (0.308) or TSFM (0.129) fine-tuned alone; it also beats specialized time-series models on 3 public benchmarks.

Time-JEPA: Grounding Video World Models with Physical Time Series *Co-first author*

Motivation World models are trained mainly on video, but the real world also has physical quantities over time (speeds, joint angles, forces) and the laws behind them. Physical laws rest on precise measurement, so world models need precise, dynamic representations of these quantities, which video-only training, good at semantics, rarely learns. Goal: bring physical time series into world-model training.

Approach Fine-tunes V-JEPA 2.1 with temporally aligned physical time series as auxiliary supervision (one encoder for three domains; video-only evaluation). Across driving, robot manipulation and human motion, versus video-only fine-tuning on the same data, it fits physical laws better: under unseen conditions, kinematic-reconstruction trajectory error drops from 4.150 m to 1.858 m; in distribution, multi-step future-state R^2 rises from 0.296 to 0.550; current-state decoding improves under unseen conditions and unseen state variables; it is more sensitive to physical anomalies (GRASP +2.2 points) and also works with the generative world model iVideoGPT.

My role Proposed the idea; designed the overall and core methods, chose the datasets and set the research hypotheses; supervised junior students who ran the experiments.

Apple, Siri China R&D Jan. 2024 – Oct. 2024

Research Intern, speech generation

BELLE: Bayesian Speech Synthesizers Can Learn from Multiple Teachers *First author* arXiv 2025

Motivation TTS is one-to-many: the same text has many valid readings whose spread depends on the input, yet most models leave this spread implicit or fix it. Goal: model it explicitly, though estimating variance needs several recordings per text.

Approach Evidential deep learning puts a Normal-Inverse-Gamma prior on each mel frame's mean and variance, predicted from the input, keeping MELLE's architecture and single-pass sampling. Built two parallel corpora: real recordings (ParallelVox, 881 h) and TTS-synthesized speech (LS+6TTS, 4,817 h). Under the same conditions it beats MELLE: zero-shot cross-sentence WER-H 2.99% vs. 3.83%, SMOS 4.13 vs. 3.80; streaming version: median first-packet latency 232 ms, RTF 0.31.

Other Research

SWIM: EEG-Based Auditory Spatial Attention Decoding with a Short-Window CNN and Mamba *First author* **IEEE SLT 2024**

Motivation Decode from EEG whether a listener attends to the left or right speaker, without clean speech, for neuro-steered hearing aids; needs fast short-window response plus long-term context.

Approach A short-window CNN with augmentation, multitask training and model combination reaches 84.9% accuracy (leave-one-speaker-out, KUL; previous best 80.0%); adding Mamba for long-term history reaches 86.2%, a 31.0% relative error reduction over the previous best.

Bayesian WeakS-to-Strong from Text Classification to Generation *Co-first author* **ICLR 2025**

Motivation Once AI surpasses humans, how can humans supervise it? We used several small models to simulate humans with diverse opinions and a larger model to simulate superintelligence, studying how multiple weak supervisors can supervise a strong model.

Approach Bayesian evidential learning estimates the confidence of the weak supervisors' labels to guide the strong model, extended to generation with conservative DPO; performance gap recovered of 0.828 on SciQ classification (best baseline 0.768).

My role Proposed the core idea; implemented the method and ran the classification experiments as the first proof of feasibility; collaborators extended it to generation.

TVPara: Paralinguistic Captioning of Expressive TV-Series Speech *First author* arXiv 2026

Motivation Audio LLMs miss paralinguistics where strong emotion meets complex acoustics; existing corpora are mostly studio speech without free-text descriptions.

Approach Built TVPara from TV series via audio-LLM pre-annotation and human correction (single-utterance, about 492 h; multi-utterance, about 213 h, describing changes between adjacent utterances). Fine-tuned open-source audio LLMs improve paralinguistic understanding by 31.26% on average (relative, vs. before fine-tuning) and generalize to unseen series.

Publications

** equal contribution*

[1] **Ziyang Zhang***, S. Li*, Y. Wang, Z. Cui, B. Zhou, W. Wu, C. Zhang. MMTA: Benchmarking Multimodal Temporal Analysis with Time Series, Text, and Vision. **NeurIPS 2026** (Evaluations & Datasets Track). arXiv

[2] W. Wu, **Ziyang Zhang**, L. Liu, X. Xu, et al., C. Zhang. SciTS: Scientific Time Series Understanding and Generation with LLMs. **ICLR 2026**. paper · code

[3] Z. Cui*, **Ziyang Zhang***, G. Sun, W. Wu, C. Zhang. Bayesian WeakS-to-Strong from Text Classification to Generation. **ICLR 2025**. paper

[4] **Ziyang Zhang**, A. Thwaites, A. Woolgar, B. Moore, C. Zhang. SWIM: Short-Window CNN Integrated with Mamba for EEG-Based Auditory Spatial Attention Decoding. **IEEE SLT 2024**. arXiv

[5] **Ziyang Zhang**, S. Li, W. Wu, C. Zhang. TimeOmni-o1: Bidirectionally Coupling Time Series Foundation Models and LLMs for Multimodal Reasoning and Forecasting. Preprint, 2026. OpenReview

[6] S. Li*, **Ziyang Zhang***, W. Wu, M. Liu, C. Zhang. Time-JEPA: Grounding Video World Models with Physical Time Series. Preprint, 2026. OpenReview

[7] **Ziyang Zhang**, W. Yu, W. Wu, X. Xu, C. Zhang. TVPara: Paralinguistic Captioning of Expressive TV-Series Speech in Complex Acoustic Environments. arXiv 2026. arXiv

[8] **Ziyang Zhang**, Y. Gao, X. Xu, B. Li, W. Wu, C. Zhang. Bayesian Speech Synthesizers Can Learn from Multiple Teachers. arXiv 2025. arXiv · demo · code

[9] Intern-S2-Preview Team (incl. **Ziyang Zhang**). Intern-S2-Preview: Scientific Agentic Foundation Model. Technical report, 2026. arXiv

[10] Intern-S1 Team (incl. **Ziyang Zhang**). Intern-S1: A Scientific Multimodal Foundation Model. Technical report, 2025. arXiv